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ENGR 095

Circuits Reflection

In this module, we learned about microcontrollers and how they can be programmed to control other devices. We began with a simple introduction to circuits by creating a butterfly-shaped LED circuit using copper tape and a battery. Then, we learned about github repositories and how they allow engineers to work together. We used this knowledge to create a portfolio website and a space to showcase our Fyre assignments with github and copilot. We then walked through the process of reading and responding to a design brief, learning about the differences between flow charts and system diagrams and isolating the necessary inputs, outputs, and constraints. Using this information, we created a concept for a security system using an Arduino microcontroller and some electrical components. We also learned about microcontrollers and experimented with programming them using micropython. We used the equipment in Keysight Lab to experiment with different types of circuits, and eventually implemented the system we designed on a breadboard using a button, a light sensor, and some LED and resistors. We uploaded the code for all of these projects to our github repository, to ensure that we created documentation of our work.

Throughout this module, I learned a lot about how electrical and computer engineers look at a design brief and analyze it to create the best solution. I really love the feeling of creating a circuit that works, and I always relish the opportunity to learn more about them. I appreciated how this section of the module builds upon the last section. Having learned about the materials

that go into electrical components and sensors, we are now able to utilize these components in more complex systems.